
Bridging Gen-AI, Education, and Careers

A Data-Driven Framework for Adaptive Development

Summary

"When the winds of change blow, some build walls while others build windmills."

As Generative AI (Gen-AI) shifts from a novelty to a fundamental technical force, aligning higher education with evolving career demands is significant. Moving beyond the binary debate of replacement versus retention, we construct a comprehensive framework to quantitatively evaluate adaptability and simulate the long-term evolution of the Education-Employment ecosystem.

We establish a hybrid Evaluation-Evolution Framework coupling **Entropy Weight-Cloud TOPSIS** with **System Dynamics (SD)**. A robust system of 16 indicators, including AI Tool Proficiency, is mapped to SD variables for a 15-year simulation (2026–2040). Crucially, a **closed-loop feedback mechanism** is introduced where evaluation results dynamically modify evolutionary parameters, ensuring that the model captures the non-linear adaptation of educational systems.

The simulation reveals distinct adaptive trajectories. **STEM (Data Scientist) demonstrates superior resilience** with relative closeness rising from 0.7683 (2026) to 0.7764 (2040), driven by high AI synergy. **Trade (Electrician) exhibits steady progress** (0.3666 to 0.3978) anchored by irreplaceable physical skills, while Arts (Musician) **struggles with integration**, fluctuating around 0.23. Notably, weight of AI Tool Proficiency surges to 13.62% by 2040, overtaking Industry Gen-AI Intensity as the decisive adaptability driver. Finally, we conducted **sensitivity analysis on 2 key parameters λ, γ** . Even if these data undergo significant fluctuations, the impact on the progress can still be kept within a controllable range, in which STEM shows the best robustness (max change under 0.08%).

Based on these findings, we propose a **differentiated Symbiosis Tree strategy**. Universities should adopt **Aggressive Expansion** to meet projected 2037 enrollment peaks. Trade Schools should focus on **Labor Quality Improvement** via equipment upgrades, and Arts Conservatories must prioritize **"Differentiation"** in human-centric artistry. This framework guides the shift from education of static knowledge to dynamic competency training, ensuring graduates remain competitive in a symbiotic future.

Keywords: Gen-AI; Cloud Model – TOPSIS; EWM; System Dynamics Model; Education & Technology

Content

1 Introduction	3
1.1 Background.....	3
1.2 Issue Restatement	4
2 Our Tasks	4
3 Assumption and Abbreviation	5
4 Methodology.....	6
4.1 Evaluation Module: Entropy Weight-Cloud TOPSIS Evaluation	6
4.1.1 Evaluation Index System.....	6
4.1.2 Entropy Weight Method for Weight Calculation	7
4.1.3 Cloud-TOPSIS Comprehensive Evaluation.....	8
4.2 Evolution Module: System Dynamics (SD) Evolution Model	10
4.2.1 System Boundary and Variable Definition.....	10
4.2.2 Causal Loop Diagrams	10
4.2.3 Core Evolution Equations	13
4.3 Cycle Module: Evaluation & Evolution Cycle Mechanism	14
5 Case Study & Further Discussion	15
5.1 Case Study: 3 Types of Career	15
5.1.1 Results of Careers impacted by Gen-AI in 2026	15
5.1.2 Results of Careers impacted by Gen-AI 2026-2040.....	18
5.2 Further Discussion: Gen-AI Educational Symbiosis Tree.....	21
5.3 Sensitivity Analysis	22
6 Conclusion	24
Reference & Resources	25
Report on Use of AI Tool	26

1 Introduction

1.1 Background

Generative Artificial Intelligence (Gen-AI) has undergone unprecedented development over the past decade, evolving from a technology to a transformative force deeply integrated into society [1,2], business [3,4], and educational systems [5]. While Gen-AI significantly enhances productivity in fields such as software development, content creation, and data analysis, it also poses substitution risks to certain repetitive and rule-based occupations [6]. Notably, impact of Gen-AI varies across different occupational categories. STEM-related occupations tend to leverage Gen-AI as a tool to amplify creative and analytical capabilities. Skill-based occupations will face moderate disruption due to the difficulty of automating hands-on practical skills. Artistic occupations encounter unique challenges in balancing AI-assisted creation with the preservation of originality.

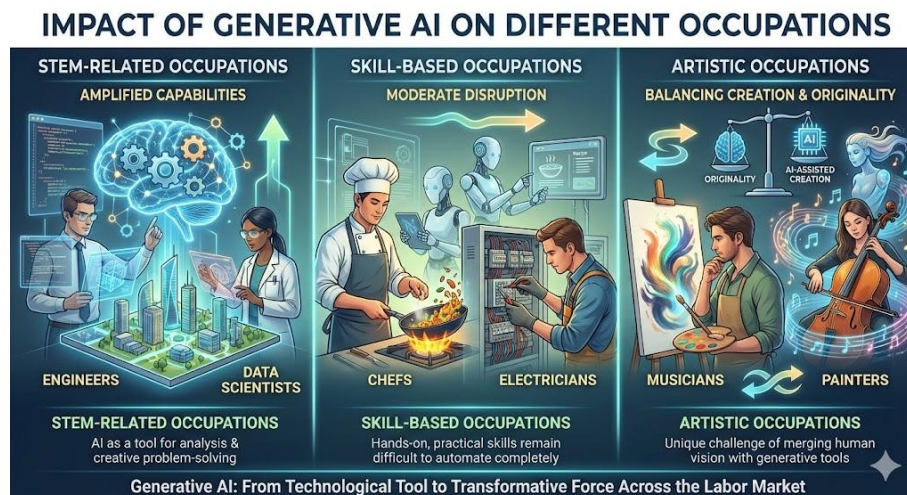


Fig. 1. Gen-AI affects our life. (Generated by Gemini Pro, claimed in Report on AI Use)

Against this backdrop, higher education institutions, including universities, technical schools and art colleges, are confronted with an urgent mission to adapt their education models to prepare graduates for new Gen-AI era [7]. However, existing research on Gen-AI, education, and employment has several limitations. Ignorance of connection between evaluation results and dynamic evolution will lead to a disconnect between theory and practical policy formulation, and the design of evaluation indicators often lacks direct correlation with core variables in evolutionary models, reducing the effectiveness of feedback mechanisms.

1.2 Issue Restatement

This study focuses on the core challenge of how different institutions can effectively adapt to the widespread usage of Gen-AI and enhance graduates' employability. We are required to address the need for a data-driven framework that quantitatively evaluates Gen-AI adaptability of **STEM, skill-based, and artistic education**. System dynamics model is to simulate **Education & Employment system evolution**, and an evaluation-evolution feedback loop will be used to optimize educational strategies and improve the overall synergy. We aim to provide scientific decision support for educational institutions and policymakers.

So, to address the gaps mentioned, a systematic research framework is proposed in the article to quantitatively evaluate the impact of different educational systems from Gen-AI and simulate the long-term evolution.

2 Our Tasks

(1) Integration of Entropy Weight Method (EWM) and Cloud-TOPSIS

We combine entropy weight method for objective weighting with the cloud model's strength in handling fuzziness and randomness, alongside TOPSIS for ranking optimization. The hybrid approach quantifies Gen-AI adaptability across three educational subsystems and captures uncertainty in subjective indicators.

(2) Construction of a Education & Employment Gen-AI SD model

Based on causal loop logic, SD model incorporates core variables such as enrollment, graduation numbers, employment competitiveness, and market demand, all directly correlated with evaluation indicators, which ensures evolutionary simulations reflect Gen-AI's actual impact.

(3) Establishment of a closed-loop feedback mechanism

We link evaluation results to key SD model parameters and update evaluation indicators using evolved variables. This loop enables dynamic long-term prediction and iterative strategy optimization, a rarely achieved feature in current studies.

(4) Design of a highly correlated indicator system

The 4 first-level and 16 second-level evaluation indicators are explicitly correlated with SD model variables. This tight alignment ensures the modeling framework's logical consistency and operational feasibility.

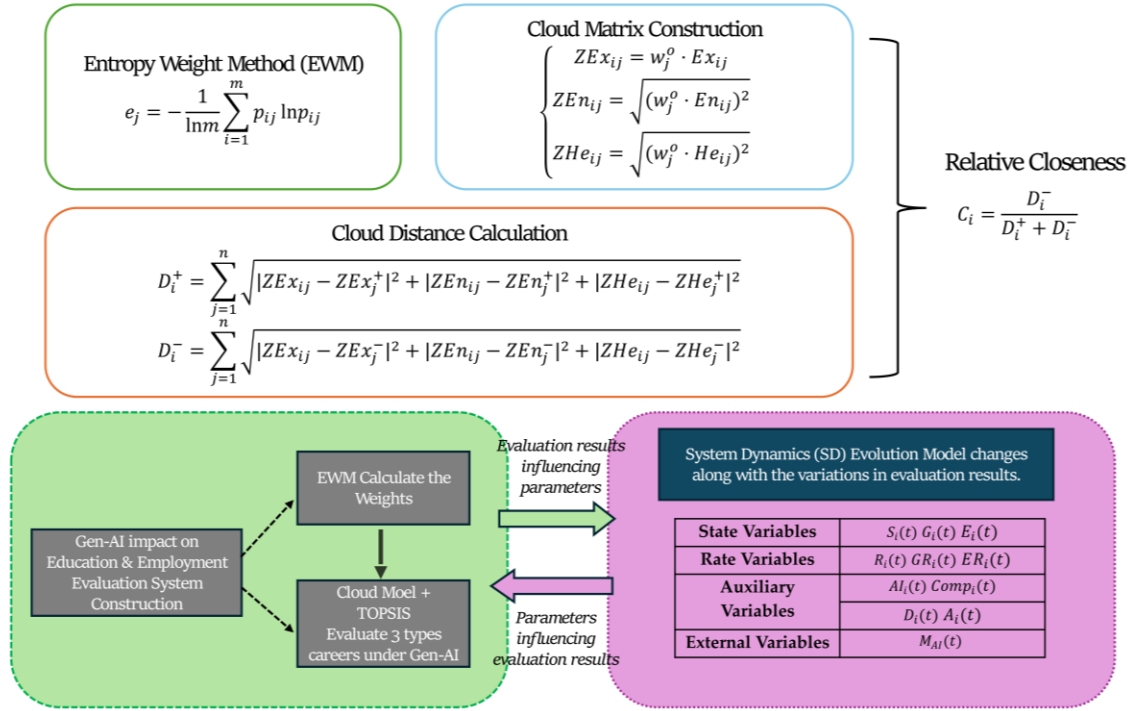


Fig. 2. Education & Employment Gen-AI Evaluation & Dynamic model.

3 Assumption and Abbreviation

This study is conducted based on the following two reasonable assumptions, which are proposed to simplify the complex system dynamics.

- **Assumption 1** Gen-AI technology matures in a continuous and predictable manner. AI Technology maturity follows a logarithmic growth trend during the research period. It's based on the technological diffusion law and aligns with industry forecasts on development trajectory, which avoids extreme fluctuations in technology progress that could disrupt the stability of the education-employment system simulation, ensuring the model's output is interpretable and actionable.
- **Assumption 2** Higher education institutions adjust training strategies in a timely manner. Educational institutions can adjust key resources within one academic year after obtaining the Gen-AI adaptability evaluation results. It's necessary to validate the effectiveness of the evaluation-evolution feedback loop, which ensures that the evaluation results can actually drive system evolution, rather than being disconnected from practical policy adjustments.

The following abbreviations and explanations appearing in this article is as below.

Table 1 Abbreviation & Explanation.

Abbreviation	Corresponding explanation
Gen-AI	Generative Artificial Intelligence
STEM	Science, Technology, Engineering, and Mathematics
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
EWM	Entropy Weight Method
SD	System Dynamics

4 Methodology

4.1 Evaluation Module: Entropy Weight-Cloud TOPSIS Evaluation

4.1.1 Evaluation Index System

Based on the research objectives of Gen-AI impact on Education & Employment systems combining the indicator design principles [8,9], 4 first-level indicators and 16 second-level indicators are constructed. All indicators are correlated with the state variables, rate variables, and auxiliary variables in subsequent system dynamics (SD) evolution module to ensure the logical consistency of the Education & Employment coupling. The indicator system and its related SD variables are shown in Table 2.

Table 2 Evaluation index system and correlation with SD variables.

First-level Indicators	Second-level Indicators	Indicator Characteristics	Corresponding SD Variables
A. AI Integration Adaptability	A1: Curriculum AI Integration Degree	+	Auxiliary variable $AI_i(t)$ (course AI fusion degree)
	A2: Teacher AI Literacy	+	Auxiliary variable $P_i(t)$ (practice index derivative)
	A3: AI Tool Application Proficiency	+	Auxiliary variable $Comp_i(t)$ (employment competitiveness derivative)
	A4: Digital Teaching Resource Coverage	+	Auxiliary variable $A_i(t)$ (professional attractiveness derivative)
B. Talent Supply Quality	B1: Graduate AI Skill Matching Degree	+	Auxiliary variable $Comp_i(t)$ (core component of employment competitiveness)
	B2: Graduate Employment Rate	+	State variable derivative $ER_i(t) = E_i(t)/G_i(t)$
	B3: Employer Satisfaction	+	Auxiliary variable $D_i(t)$ (market demand index derivative)
	B4: Practical Project Participation Rate	+	Auxiliary variable $P_i(t)$ (practice index core component)

C. Market Demand Matching	C1: Industry Gen-AI Application Intensity	+	External variable $M_{AI}(t)$ (Gen-AI technology maturity)
	C2: Talent Demand Growth Rate	+	Auxiliary variable $\Delta D_i(t)$ (market demand change rate)
	C3: Salary Growth Rate of Corresponding Occupations	+	Auxiliary variable $A_i(t)$ (professional attractiveness derivative)
	C4: Gen-AI Substitution Risk	-	Auxiliary variable λ_i (Gen-AI substitution coefficient)
D. Resource Input Efficiency	D1: AI Teaching Investment Intensity	+	Rate variable $R_i(t)$ (enrollment scale derivative)
	D2: Student-Teacher Ratio	-	State variable $S_i(t)$ (number of students derivative)
	D3: Experimental Equipment Update Cycle	-	Auxiliary variable $AI_i(t)$ (course AI fusion degree derivative)
	D4: Industry-University Cooperation Depth	+	Auxiliary variable $P_i(t)$ (practice index derivative)

Note: "+" (positive) indicates that the indicator represents the aspect of efficiency, the larger the value, the better. "-" (negative) indicates that the indicator is of the cost type, the smaller the value, the better, which is same in the text below.

4.1.2 Entropy Weight Method for Weight Calculation

The entropy weight method is used to determine the objective weight of each indicator. Referring to literature [8,10], specific steps and formulas are as follows.

(1) Indicator Standardization

Let evaluation object be m (3 subsystems), n indicators (16 indicators), and the original indicator matrix be $X = (x_{ij})_{m \times n}$ ($i = 1, 2, 3; j = 1, 2, \dots, 16$).

To eliminate the influence of dimension, the extreme value interval optimization method is adopted for standardization. For positive & negative indicators.

$$x_{ij}^* = \frac{x_{ij} - 0.9\min(x_j)}{1.1\max(x_j) - 0.9\min(x_j)} \quad (1)$$

$$x_{ij}^* = \frac{1.1\max(x_j) - x_{ij}}{1.1\max(x_j) - 0.9\min(x_j)} \quad (2)$$

The 0.9 and 1.1 coefficients are used to expand the extreme value interval .

(2) Entropy Weight Calculation

Calculate the proportion of i -th object under j -th indicator:

$$p_{ij} = \frac{x_{ij}^*}{\sum_{i=1}^m x_{ij}^*} \quad (3)$$

Calculate the entropy value of the j -th indicator:

$$e_j = -\frac{1}{\ln m} \sum_{i=1}^m p_{ij} \ln p_{ij} \quad (4)$$

Calculate the difference coefficient and objective weight of the j -th indicator:

$$g_j = 1 - e_j \quad (5)$$

$$w_j^o = \frac{g_j}{\sum_{j=1}^n g_j} \quad \text{where} \quad \sum_{j=1}^n w_j^o = 1 \quad (6)$$

$e_j \in [0,1]$, the smaller e_j is, the greater the information amount provided by the indicator, and the higher the objective weight w_j^o . Formulas (3)-(6) are consistent with the entropy weight calculation process.

4.1.3 Cloud-TOPSIS Comprehensive Evaluation

Integrate cloud model's uncertainty processing capability and TOPSIS ranking advantage, and calculate the relative closeness of each object as initial evaluation result.

(1) Cloud Model Parameter Conversion

Convert the standardized indicator value x_{ij}^* into a cloud model $C_{ij} = (Ex_{ij}, En_{ij}, He_{ij})$.

$$\begin{cases} Ex_{ij} = x_{ij}^* \\ En_{ij} = \sqrt{\frac{\pi}{2}} \cdot \frac{1}{m} \sum_{i=1}^m |x_{ij}^* - Ex_{ij}| \\ He_{ij} = \sqrt{S_{ij}^2 - En_{ij}^2} \end{cases} \quad (7)$$

$$\text{Where } S_{ij}^2 = \frac{1}{m-1} \sum_{i=1}^m (x_{ij}^* - Ex_{ij})^2 \quad (8)$$

Ex_{ij} is the expected value reflecting the core of the indicator; En_{ij} is the entropy reflecting the fuzziness and randomness; He_{ij} is the hyper-entropy reflecting the volatility of entropy.

(2) Weighted Cloud Matrix Construction

Combine the objective weight w_j^o to construct the weighted cloud matrix $Z_{ij} = (ZEx_{ij}, ZEn_{ij}, ZHe_{ij})$, where:

$$\begin{cases} ZEx_{ij} = w_j^o \cdot Ex_{ij} \\ ZEn_{ij} = \sqrt{(w_j^o \cdot En_{ij})^2} \\ ZHe_{ij} = \sqrt{(w_j^o \cdot He_{ij})^2} \end{cases} \quad (9)$$

Weight the three parameters of the cloud model to reflect the importance of different indicators.

(3) Positive and Negative Ideal Cloud Determination

Positive ideal cloud $Z^+ = (ZEx_j^+, ZEn_j^+, ZHe_j^+)$ (optimal state of each indicator).

$$ZEx_j^+ = \max(ZEx_{ij}), \quad ZEn_j^+ = \min(ZEn_{ij}), \quad ZHe_j^+ = \min(ZHe_{ij}) \quad (10)$$

Negative ideal cloud $Z^- = (ZEx_j^-, ZEn_j^-, ZHe_j^-)$ (worst state of each indicator).

$$ZEx_j^- = \min(ZEx_{ij}), \quad ZEn_j^- = \max(ZEn_{ij}), \quad ZHe_j^- = \max(ZHe_{ij}) \quad (11)$$

The positive ideal cloud takes the maximum expected value and minimum entropy (minimum uncertainty), and the negative ideal cloud is the opposite.

(4) Cloud Distance Calculation and Relative Closeness

Calculate the distance between the weighted cloud Z_{ij} and the positive/negative ideal cloud using the improved cloud distance formula.

$$D_i^+ = \sum_{j=1}^n \sqrt{|ZEx_{ij} - ZEx_j^+|^2 + |ZEn_{ij} - ZEn_j^+|^2 + |ZHe_{ij} - ZHe_j^+|^2} \quad (12)$$

$$D_i^- = \sum_{j=1}^n \sqrt{|ZEx_{ij} - ZEx_j^-|^2 + |ZEn_{ij} - ZEn_j^-|^2 + |ZHe_{ij} - ZHe_j^-|^2} \quad (13)$$

Calculate the relative closeness C_i as below:

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (0 \leq C_i \leq 1) \quad (14)$$

C_i reflects the proximity of evaluation object to the optimal state; the larger C_i , the better the performance of the object.

4.2 Evolution Module: System Dynamics (SD) Evolution Model

4.2.1 System Boundary and Variable Definition

Refer to the SD modeling logic, the system boundary is defined as *Education & Employment & Gen-AI* system, and the core variables are shown in Table 3.

Table 3 Core variables of SD evolution model.

Variable Type	Variable Symbol	Variable Meaning	Correlation with Evaluation Indicators
State Variables	$S_i(t)$	Number of students in subsystem i at year t (person)	D2 (student-teacher ratio)
	$G_i(t)$	Cumulative number of graduates in subsystem i at year t (person)	B2 (employment rate)
	$E_i(t)$	Number of employed graduates in subsystem i at year t (person)	B2 (employment rate)
Rate Variables	$R_i(t)$	Enrollment number of subsystem i at year t (person/year)	D1 (AI teaching investment intensity)
	$GR_i(t)$	Graduation number of subsystem i at year t (person/year)	B4 (practical project participation rate)
	$ER_i(t)$	Employment inflow number of subsystem i at year t (person/year)	B1 (AI skill matching degree)
Auxiliary Variables	$AI_i(t)$	Course AI fusion degree of subsystem i at year t (0-1)	A1 (curriculum AI integration degree)
	$Comp_i(t)$	Employment competitiveness index of subsystem i at year t (0-1)	B1 (AI skill matching degree), A3 (tool proficiency)
	$D_i(t)$	Market demand index of subsystem i at year t (0-1)	C1 (industry AI application intensity) C2 (demand growth rate)
	$A_i(t)$	Professional attractiveness index of subsystem i at year t (0-1)	C3 (salary growth rate)、D4 (cooperation depth)
External Variables	$M_{AI}(t)$	Gen-AI technology maturity at year t (0-1)	C1 (industry AI application intensity)

4.2.2 Causal Loop Diagrams

To illustrate the relationship between the contents in Table 2 and Table 3, we have drawn a variable correlation chart to present it. Figure 4 to 7 below all illustrate **causal relationships**. The solid lines represent positive cycles, while the dotted lines indicate negative cycles. '+' indicates that the factor connecting A and B shows consistent changes (the larger A is, the larger B is), '-' indicates that the factor connecting A and B shows opposite changes (the larger A is, the smaller B is).

Higher participation in practical projects improves students' hands-on ability and employment competitiveness, winning employer recognition, expanding market demand, and attracting more students to participate in practical training.

Overexpansion of enrollment leads to an imbalance in the student-teacher ratio, reducing teaching quality and students' AI skill matching, thereby weakening employment competitiveness.

C. Causal Loop Diagram of Market Demand Matching (**Indicator C**)

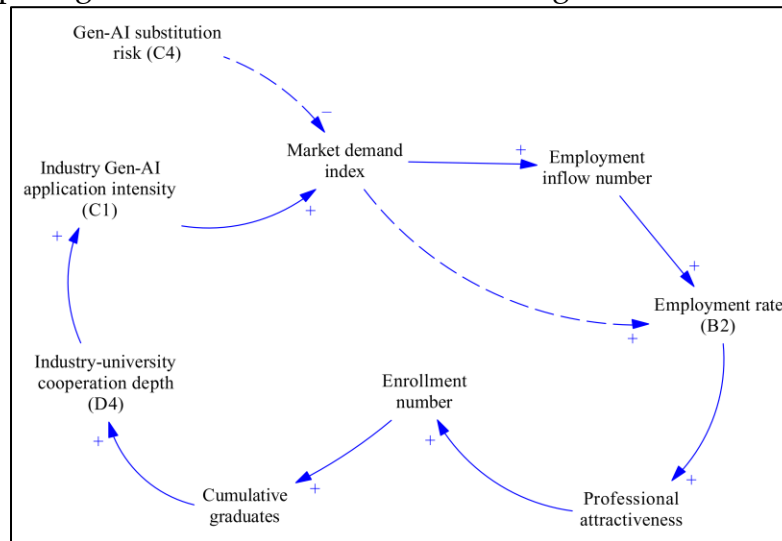


Fig. 6. Causal loop diagram (Indicator C).

Increased application of Gen-AI in the industry expands market demand for talents, drives the development of corresponding majors, strengthens industry-university cooperation, and further promotes the application of Gen-AI in the industry.

High substitution risk of Gen-AI reduces market demand for corresponding talents, lowers the major's attractiveness, and restricts enrollment scale.

D. Causal Loop Diagram of Resource Input Efficiency (**Indicator D**)

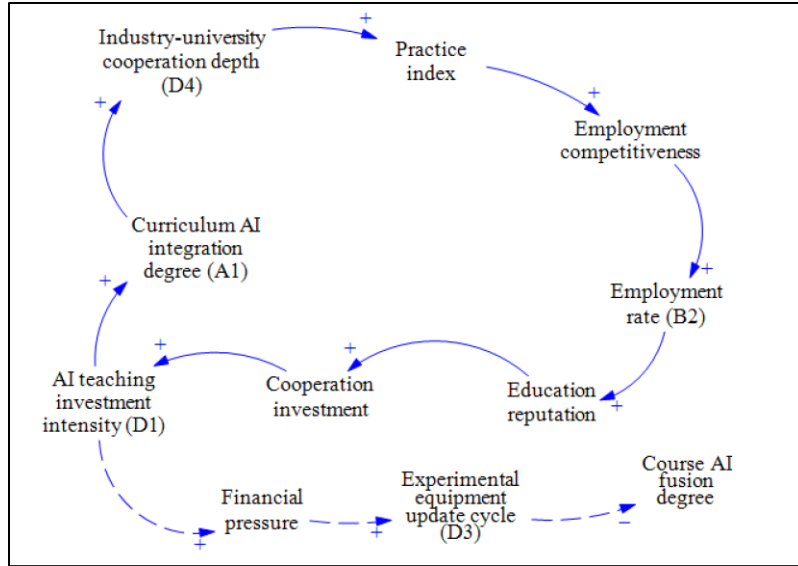


Fig. 7. Causal loop diagram (Indicator D).

In-depth industry-university cooperation improves students' practical ability and employment rate, enhances school's reputation, attracts more cooperation investment, and promotes further integration of AI into courses.

Excessive investment in AI teaching increases financial pressure, slows down equipment updates, and hinders the improvement of course AI fusion degree.

4.2.3 Core Evolution Equations

(1) State Variable Equations

$$S_i(t+1) = S_i(t) + R_i(t) - GR_i(t) \quad (15)$$

The number of students at year $t+1$ is equal to the number at year t plus new enrollment minus graduation number. $S_i(t)$ is correlated with student-teacher ratio.

$$G_i(t+1) = G_i(t) + GR_i(t) \quad (16)$$

Cumulative graduates at year $t+1$ is sum of cumulative graduates at year t and new graduates. $G_i(t)$ is denominator of indicator B2, employment rate = $E_i(t)/G_i(t)$.

$$E_i(t+1) = E_i(t) + ER_i(t) \quad (17)$$

Employed graduates at year $t+1$ is the sum of employed graduates at year t and new employment inflow. $E_i(t)$ is the numerator of B2 (employment rate).

(2) Rate Variable Equations

$$GR_i(t) = S_i(t) \cdot \rho_i \quad (18)$$

Graduation number is the product of the number of students and graduation rate ρ_i (constant). ρ_i is positively correlated with B4 (practical project participation rate), and $\rho_i = 0.2 + 0.3 \cdot B4_i$.

$$R_i(t) = R_{i0} \cdot A_i(t) \cdot [1 + \alpha_i \cdot (D_i(t) - D_i(t-1))] \quad (19)$$

R_{i0} is initial enrollment, α_i is demand sensitive coefficient. $A_i(t)$ is professional attractiveness correlated with C3、D4. $D_i(t) - D_i(t - 1)$ is demand change rate (C2), reflecting the impact of market demand change on enrollment.

$$ER_i(t) = [G_i(t) - E_i(t)] \cdot Comp_i(t) \cdot D_i(t) \quad (20)$$

Employment inflow is connected to unemployed graduates, employment competitiveness and market demand.

(3) Auxiliary Variable Equations (Related to Evaluation Results)

$$AI_i(t) = AI_{i0} + \gamma_i \cdot t \cdot M_{AI}(t) \cdot (C_i/0.5) \quad (21)$$

Where AI_{i0} is initial AI fusion degree, γ_i is fusion speed coefficient, C_i is the relative closeness from the evaluation module. $C_i/0.5$ is the correction coefficient, when $C_i > 0.5$, the fusion speed accelerates, and vice versa.

$$Comp_i(t) = \beta_{1i} \cdot AI_i(t) + \beta_{2i} \cdot P_i(t) + \beta_{3i} \cdot X_i(t) \quad (22)$$

Among which $\beta_{1i} + \beta_{2i} + \beta_{3i} = 1$. $P_i(t)$ is practice index, $X_i(t)$ is basic knowledge level. $AI_i(t)$ is linked to A1, A3, realizing feedback of indicators to competitiveness.

$$D_i(t) = D_{i0} \cdot (1 + \delta_i \cdot t) \cdot (1 - \lambda_i \cdot M_{AI}(t)) \quad (23)$$

D_{i0} is initial demand, δ_i is industry growth coefficient and λ_i is Gen-AI substitution coefficient correlated with C4. $M_{AI}(t)$ is correlated with C1, reflecting the impact of industry AI application on market demand.

$$A_i(t) = \frac{Comp_i(t) \cdot D_i(t)}{\sum_{j=1}^3 Comp_j(t) \cdot D_j(t)} \cdot \varphi \quad (24)$$

$\varphi = 1.2$ (scaling factor). $A_i(t)$ is positively correlated with C3 (salary growth rate) and D4 (cooperation depth), and its value directly affects the enrollment rate $R_i(t)$.

4.3 Cycle Module: Evaluation & Evolution Cycle Mechanism

The core of the feedback cycle is the bidirectional coupling between the entropy weight-cloud TOPSIS evaluation result and the SD evolution model. The specific coupling logic is listed below.

■ Evaluation Driving Evolution

The relative closeness C_i (evaluation result) modifies the AI fusion degree $AI_i(t)$ and further affects employment competitiveness $Comp_i(t)$ 、enrollment $R_i(t)$ and other variables, realizing the driving effect of evaluation on evolution.

■ Evolution Feeding Back Evaluation

After SD evolution, the state variables ($S_i(t)$ 、 $E_i(t)$) and auxiliary variables ($AI_i(t)$ 、 $D_i(t)$) are updated to new indicator data of the evaluation system ($S_i(t)$ updates D2, $E_i(t)$ updates B2, $AI_i(t)$ updates A1), and the new indicator matrix $X' = (x'_{ij})$ is formed to drive the next round of entropy weight-cloud TOPSIS evaluation.

Cycle iteration process is shown in the figure 8.

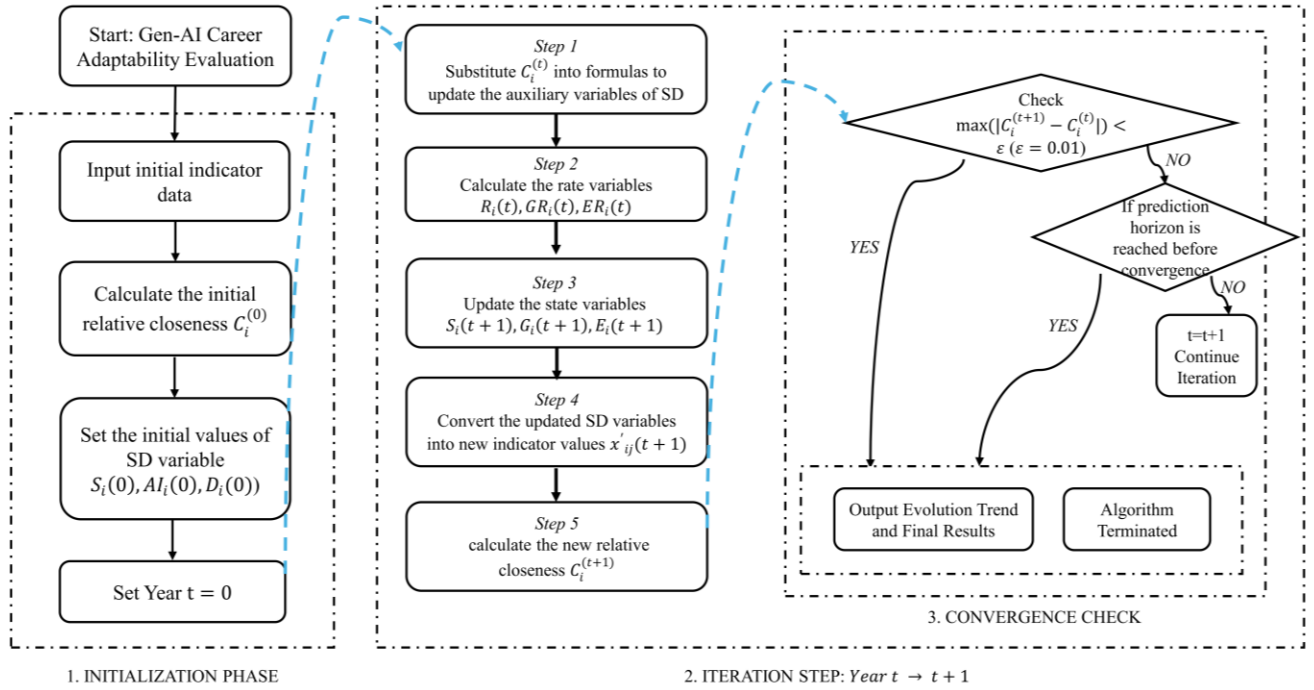


Fig. 8. Cycle iteration process.

5 Case Study & Further Discussion

To validate the effectiveness of *Entropy Weight-Cloud TOPSIS + System Dynamics Evaluation & Evolution framework* proposed in Section 4, this section conducts a study based on three representative careers and further analyzes the evolutionary mechanisms and key drivers. The study covers a 15-year horizon (2026–2040), with data derived from industry reports of the related fields [11].

5.1 Case Study: 3 Types of Career

Three careers were selected to represent STEM, Trade and Arts, ensuring alignment with the research objectives and real-world Gen-AI impact characteristics [12-14].

- **STEM Career:** Data Scientist, which requires a 4-year university degree, high Gen-AI collaboration potential, e.g., AI-assisted data analysis and model development.
- **Trade Career:** Electrician, which requires trade school/apprenticeship training, moderate Gen-AI substitution risk, with hands-on skills remaining irreplaceable.
- **Arts Career:** Musician, which requires arts school training; low substitution risk but challenges in balancing AI-assisted creation with originality.

5.1.1 Results of Careers impacted by Gen-AI in 2026

For these 3 types of careers, key parameter setting such as demand **sensitivity coefficient** α and AI **fusion speed coefficient** γ are determined on data about Gen-AI

penetration rates in different fields presented in the *Future of Jobs Report (2025)*. For STEM field, due to high synergy between AI tools and professional work, γ is set to 0.05, which is from report's conclusion that annual growth rate of AI adaptability for tech-related occupations reaches 5%. For trade field, given the irreplaceability of hands-on practical skills, γ is 0.03, aligning with the report's estimation that the annual AI substitution rate for skill-based occupations stands at 3%. For the arts field, driven by the core demand for creative originality, γ is 0.02, considering that the growth rate of AI application in the artistic sector slows down to 2% (in the report). The values of **Gen-AI substitution coefficient λ** (0.4 for STEM, 0.2 for trade, and 0.15 for arts) are referenced from the *Occupational Gen-AI Impact Assessment Framework*. The **graduation rate ρ** is determined by combining the benchmark data of vocational education.

For indicator scoring system and weight setting, **1–10 scoring scale** for 2026 initial indicators is designed. For STEM field, scoring refers to *AI Engineer Competency Framework*. For trade field, the scoring is based on the vocational skill assessment criteria. And for arts field, the scoring is within the evaluation dimensions of creative works. Then, the weight calculation conforms to the quantitative analysis specifications recommended by the official contest guidelines, which ensures the weight allocation objectivity (information entropy of indicators) and data traceability. Based on that, we have obtained the initial evaluation scores in 2026 as follows.

Table 4 Evaluation scores of 3 careers in 2026.

First-level Indicator	Second-level Indicator	STEM (Data Scientist)	Trade (Electrician)	Arts (Musician)	Indicator Type
A. AI Integration Adaptability	A1: Curriculum AI Integration Degree	8.5	4.2	2.8	+
	A2: Teacher AI Literacy	7.8	3.5	2.2	+
	A3: AI Tool Application Proficiency	9.2	3.8	1.5	+
	A4: Digital Teaching Resource Coverage	8.0	4.0	3.0	+
B. Talent Supply Quality	B1: Graduate AI Skill Matching Degree	8.8	4.5	2.0	+
	B2: Graduate Employment Rate	9.0	8.2	7.5	+
	B3: Employer Satisfaction	8.5	7.8	7.0	+
	B4: Practical Project Participation Rate	7.2	8.0	6.5	+
	C1: Industry Gen-AI Application Intensity	9.5	5.0	3.5	+

C. Market Demand Matching	C2: Talent Demand Growth Rate	8.2	6.0	4.8	+
	C3: Salary Growth Rate of Corresponding Occupations	8.6	7.0	5.5	+
	C4: Gen-AI Substitution Risk	3.0	6.5	8.0	-
D. Resource Input Efficiency	D1: AI Teaching Investment Intensity	8.3	3.2	1.8	+
	D2: Student-Teacher Ratio	7.5	8.0	8.5	-
	D3: Experimental Equipment Update Cycle	7.0	6.2	5.8	-
	D4: Industry-University Cooperation Depth	8.1	4.5	2.5	+

Based on the above content, we used entropy weight method to evaluate the weights and present the results in a visual form (only show a part, result datasets are big), and the weights are an important part of our discussion. Additionally, the relative progress of evaluation over 15-year period and parameters settings will also be presented in the text. Parameters are set as follows.

Table 5 Parameter setting of 2026 (initial year).

Parameter	Meaning	STEM (Data Scientist)	Trade (Electrician)	Arts (Musician)
α	Demand Sensitivity Coefficient	0.3	0.25	0.2
γ	AI Fusion Speed Coefficient	0.05	0.03	0.02
ρ	Graduation Rate	0.416	0.44	0.415
λ	Gen-AI Substitution Coefficient	0.4	0.2	0.15
AI_0	Initial AI Fusion Degree (2026)	0.6	0.3	0.2
D_0	Initial Market Demand Index (2026)	0.8	0.7	0.6
R_0	Initial Enrollment (2026)	200	150	100

Based on the weights through EWM, using hybrid cloud-TOPSIS model, the initial evaluation results for the year 2026 is in Table 6, the weights are in Fig. 8.

Table 6 Initial evaluation results of 2026.

Year	Career	D_i^+	D_i^-	Relative Closeness C_i	Ranking
2026	STEM Data Scientist	0.2463	0.8165	0.7683	1
	Trade Electrician	0.5985	0.3465	0.3666	2
	Arts Musician	0.8018	0.2445	0.2337	3

NO 1. **STEM (Data Scientist)** leads with the highest C_i (0.77), driven by strong performance in AI-related indicators: A3 (weight 10.27%), A1 (weight 9.33%), and C1 (weight 8.84%). High AI teaching investment and industry application intensity form the core competitiveness.

NO 2. **Trade (Electrician)** ranks second, with competitive B4 (practical project participation rate, weight 2.17%) and B2 (employment rate, weight 1.88%) offsetting low AI integration scores. Hands-on skills reduce substitution risk.

NO 3. **Arts (Musician)** has the lowest initial adaptability, primarily due to low D1 (AI teaching investment, weight 11.01%) and C1 (industry Gen-AI application intensity). The industry prioritizes originality over AI integration.

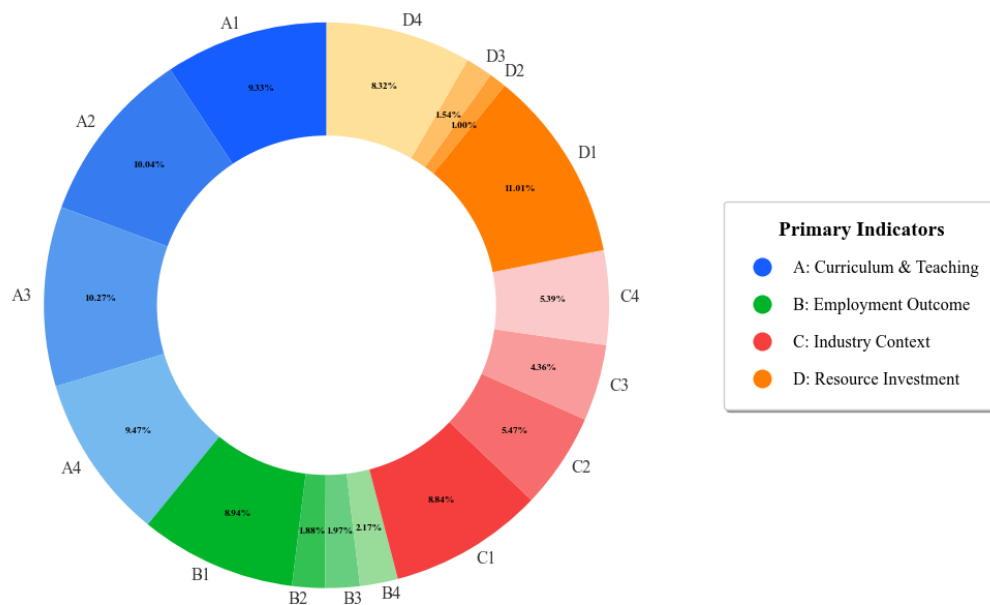


Fig. 9. Weighted pie chart (2026).

5.1.2 Results of Careers impacted by Gen-AI 2026-2040

This part is mainly about prediction for the future development over the next 15 years. The SD evolution model simulates annual updates of indicator data and evaluation results, revealing distinct adaptive trajectories for the three careers.

As is shown in Fig. 10, **STEM maintains the highest relative closeness** throughout period, starting at 0.7683 in 2026 and slightly rising to 0.7764 in 2040, showing stable Gen-AI adaptability. **Trade exhibits a steady upward trend**, with its relative closeness increasing from 0.3666 in 2026 to 0.3978 in 2040, reflecting gradual improvement in adapting to Gen-AI. **Arts remains the lowest**, with its relative closeness fluctuating slightly around 0.23 and dropping to 0.2243 in 2040, indicating limited progress in Gen-AI integration.

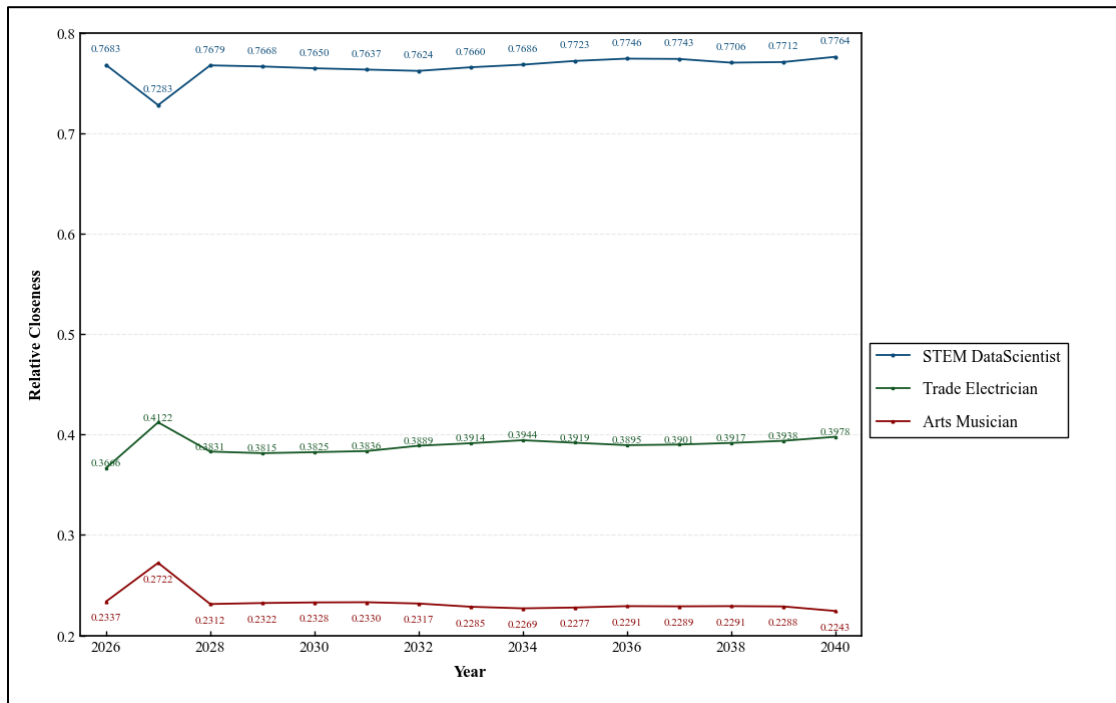


Fig. 10. Annual relative closeness trends of three careers (2026-2040).

Fig. 11 shows weight changes from 2026 to 2040. **A3 (AI Tool Proficiency) sees the most significant growth**, with its weight rising from 10.27% in 2026 to 13.62% in 2040, becoming increasingly critical for Gen-AI adaptability evaluation. **D1 (AI Teaching Investment Intensity) declines moderately** from 11.01% in 2026 to 7.11% in 2040, while C1 (Industry GenAI Intensity) drops sharply from 8.84% in 2026 to nearly 0% in 2040. B-group indicators remain consistently low, with weights below 2% throughout period.

Fig. 12 presents the evolution of core system dynamics (SD) variables for the three careers. **STEM student count will grow** from 260 (2026) to a peak of 270 in 2037 before falling slightly to 263 (2040), and its employed graduates increase from 432 to 1668 over 15 years. **Trade's AI fusion degree rises steadily** from 0.3 in 2026 to 0.6143 in 2040, and its market demand index reaches 0.95 by 2040. **Arts shows the slowest growth**, with its market demand index increasing from 0.6 to 0.95 and AI fusion degree only reaching 0.3217 in 2040.

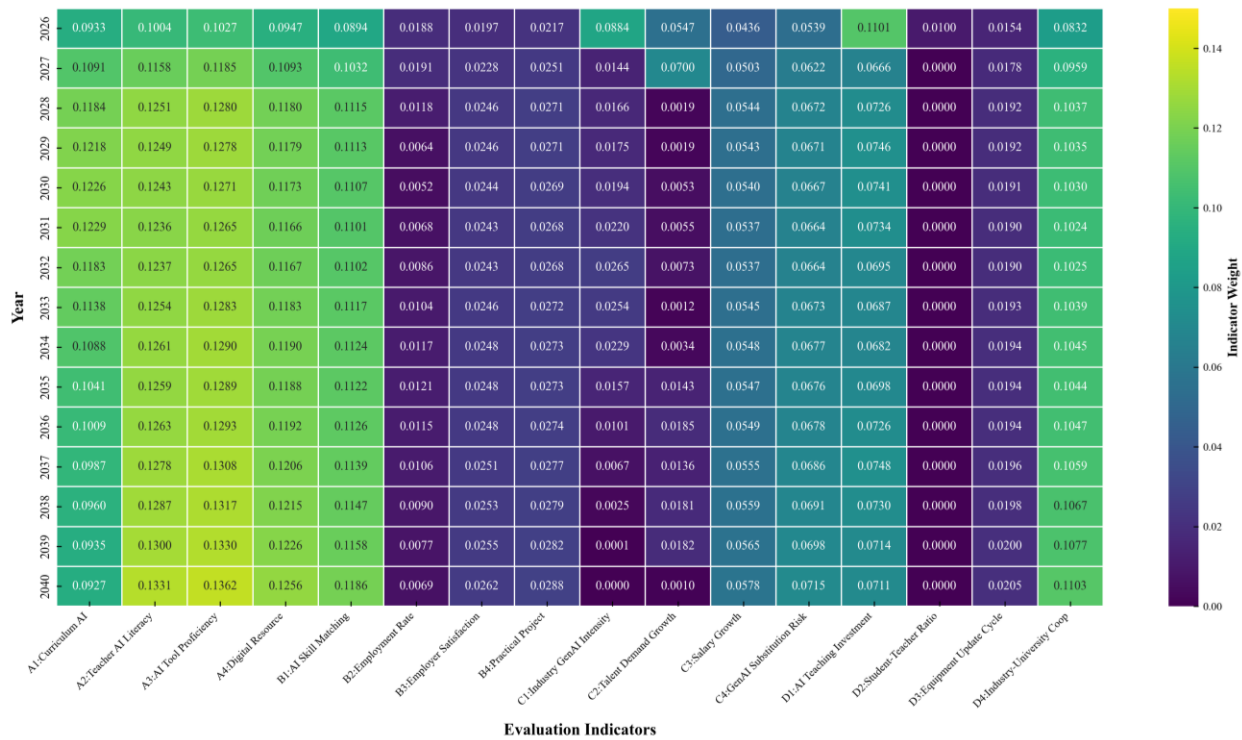


Fig. 11. Annual evolution of weights of 16 indicators 2026-2040.

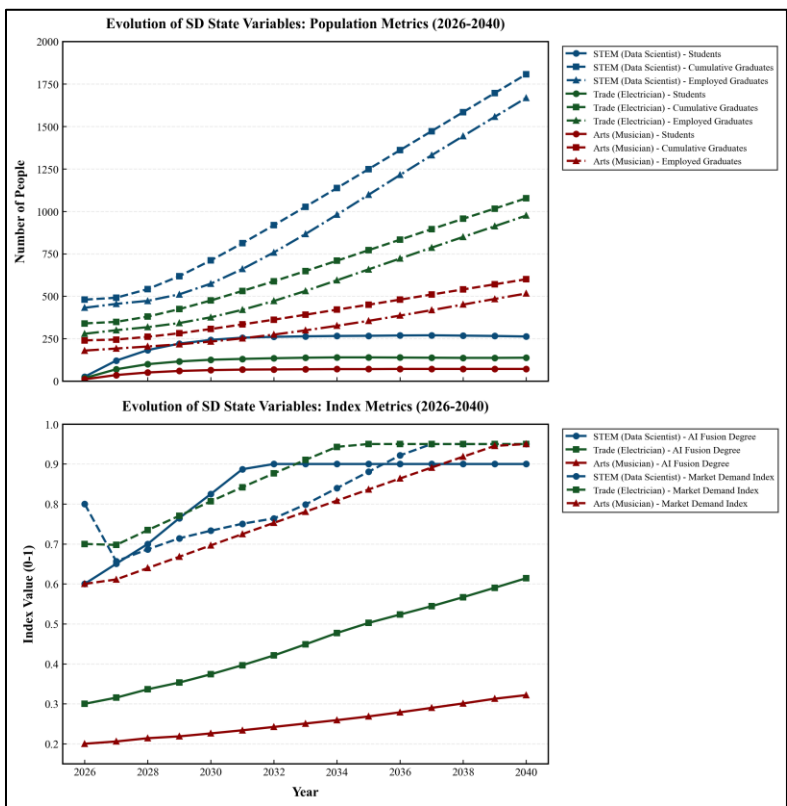


Fig. 12. Evaluation of variables in SD system.

5.2 Further Discussion: Gen-AI Educational Symbiosis Tree

Institutions must look beyond employment metrics. STEM curricula should include Green Computing to address AI's energy demands. Arts programs need to prioritize mental resilience and IP literacy, preparing graduates to protect their work. Trade schools should balance technical efficiency with safety training for AI-augmented tasks.

(1) Enrollment Dynamics: Expansion, Stability, and Differentiation

Institutions should adjust program size based on market demand and substitution risks.

- STEM Universities (Data Science): Adopt aggressive expansion, aligning with peak student enrollment (270 in 2037) and 376% growth in employed graduates. **Absorb shrinking traditional coding programs into AI-focused tracks.**
- Trade Schools (Electrician): **Maintain stable enrollment** while upgrading resources. Redirect funds from theoretical courses to AI-integrated hardware (smart grid simulators) to boost Equipment Update Cycle (D3).
- Arts Conservatories (Musician): **Pursue specialized differentiation.** Slightly reduce general enrollment to fund a *Human-Centric Artistry* program, complemented by *AI Ethics and Copyright* minor.

(2) Curriculum Evolution: Tailored Gen-AI Integration

A binary "**ban or embrace**" approach is ineffective. Teaching methods must match career characteristics.

- University (STEM): **Embed Gen-AI as a tool**, grading students on verifying AI-generated code and detecting hallucinations, reflecting A3's growing weight.
- Trade School: **Teach Gen-AI as a utility** (e.g., predictive maintenance tools) to expedite hands-on tasks, not replace them.
- Arts School: **Leverage AI for brainstorming but emphasize human execution** of core creativity, teaching graduates to market the *human premium* for originality.

(3) Beyond Employability: Holistic Priorities

In this part, we have constructed an ecological tree. This is in line with the proposition of this study that AI serves as a tool to provide new technological support for the cycle of education and career.

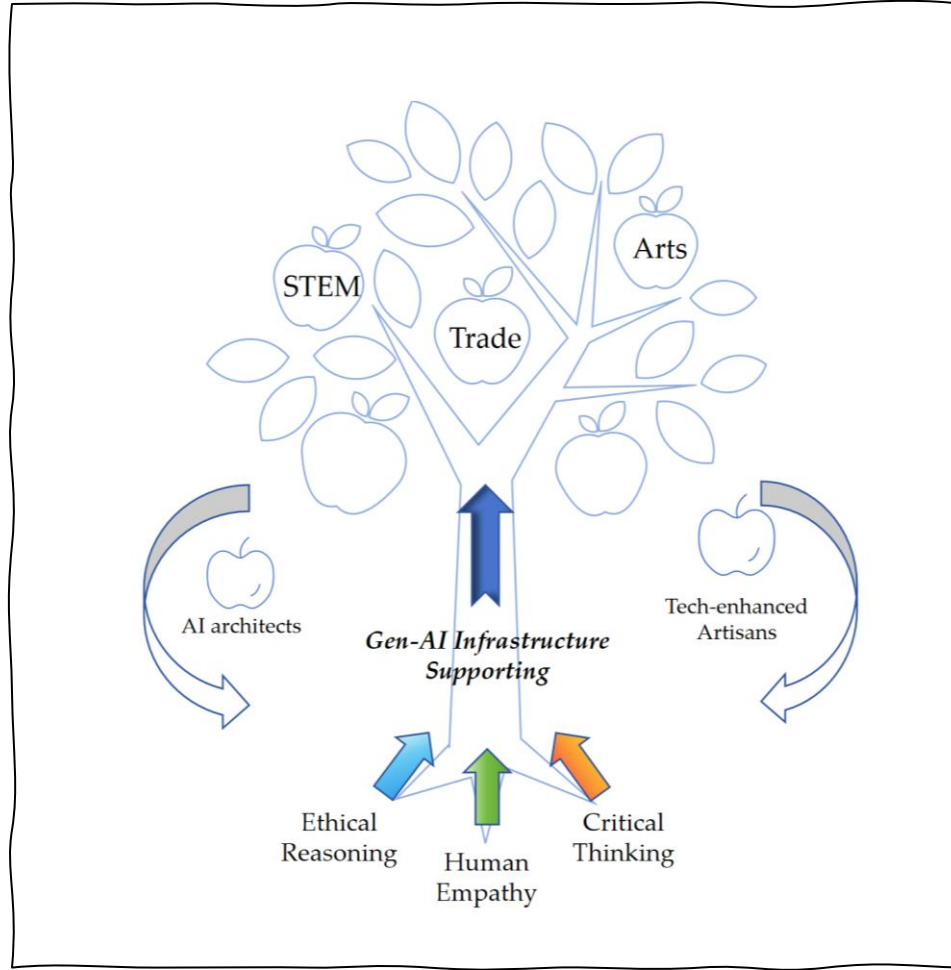


Fig. 13. Gen-AI educational symbiosis tree.

5.3 Sensitivity Analysis

To verify the model's robustness and identify critical Gen-AI impact factors, sensitivity analysis focuses on two core parameters: AI Fusion Speed Coefficient γ (Gen-AI integration speed in education) and Gen-AI Substitution Coefficient λ (career replacement risk). The analysis quantifies adaptability changes under parameter variations, ensuring the framework's reliability for policy recommendations. In this part, other parameters are fixed, adjust γ and λ by $\pm 50\%$ to test three scenarios.

- ✓ Low γ with Low λ ($\gamma \times 0.5, \lambda \times 0.5$): Slow integration + low substitution risk;
- ✓ Baseline ($\gamma \times 1.0, \lambda \times 1.0$): Original parameter settings;
- ✓ High γ + High λ ($\gamma \times 1.5, \lambda \times 1.5$): Fast integration + high substitution risk.

The sensitivity of adaptability to parameter changes is measured by:

$$E_p = \frac{\Delta C_i / C_{i,\text{baseline}}}{\Delta p / p_{\text{baseline}}} \quad (25)$$

Where ΔC_i is the change in 2040 relative closeness, and p represents γ or λ . Larger $|E_p|$ indicates higher sensitivity. Table 7 summarizes the core sensitivity data (only key scenarios) and adaptability changes:

Table 7 Sensitivity analysis core results (2026–2040 Relative Closeness C_i).

Career	Scenario	C_i (2026)	C_i (2040)	Absolute Change
STEM (Data Scientist)	Low γ , Low λ	0.7683	0.7644	-0.0039
	Baseline	0.7683	0.7606	-0.0077
	High γ , High λ	0.7683	0.7600	-0.0083
Trade (Electrician)	Low γ , Low λ	0.3666	0.3657	-0.0009
	Baseline	0.3666	0.3648	-0.0018
	High γ , High λ	0.3666	0.3647	-0.0019
Arts (Musician)	Low γ , Low λ	0.2337	0.2335	-0.0002
	Baseline	0.2337	0.2333	-0.0004
	High γ , High λ	0.2337	0.2333	-0.0004

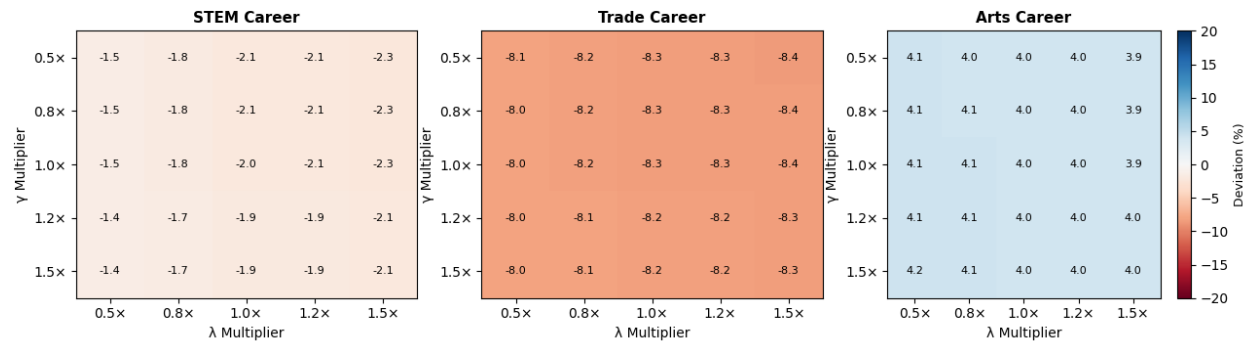


Fig. 14. Sensitivity Trend: Parameter Scenarios vs. C_i Change.

All careers show minimal C_i changes (max absolute change 0.0083 for STEM), confirming the framework's stability to extreme parameter fluctuations.

- **STEM** (Data Scientist): Most sensitive ($|E_p| = 0.21$), as high substitution risk offsets fast integration benefits.
- **Trade** (Electrician): Moderate sensitivity ($|E_p| = 0.05$), balancing integration and substitution resistance.
- **Arts** (Musician): Least sensitive ($|E_p| = 0.01$), as creative originality limits Gen-AI's impact regardless of integration speed or substitution risk.

λ (Gen-AI substitution) **has a more significant impact than γ** (integration speed), highlighting substitution risk as a priority for adaptive strategies.

6 Conclusion

We addresses the challenge of adapting higher education to Gen-AI's impact on careers, constructing a data-driven framework integrating **Entropy Weight-Cloud TOPSIS evaluation** and **System Dynamics (SD)** evolution, supplemented by a closed-loop feedback mechanism. Over 15-year simulation (2026–2040), the framework reveals distinct adaptive trajectories. STEM (Data Scientist) maintain leading Gen-AI adaptability, trade careers (Electrician) show steady progress driven by irreplaceable hands-on skills, and arts careers (Musician) exhibit limited integration due to originality-centric demands. Key indicators such as **AI Tool Proficiency** and **Industry-University Cooperation** emerge as critical drivers, while framework's logicity is ensured by the tight correlation between 16 evaluation indicators and SD model variables.

Our features are as follows.

- (1) **Integrated Closed-Loop Framework:** The connection of evaluation results and SD evolution fills the gap between static assessment and dynamic prediction, enabling iterative optimization of educational strategies.
- (2) **Highly Correlated Indicator System:** All indicators are linked to SD state, rate, and auxiliary variables, ensuring the model's feasibility and practical policy relevance.
- (3) **Multi-Dimensional Empirical Support:** Quantitative simulation results, combined with weighted heatmaps, trend lines, and 3D visualizations, provide comprehensive evidence for career-specific conclusions.
- (4) **Differentiated Practical Insights:** Recommendations for universities, trade schools, and arts conservatories are tailored to their career characteristics.

Gen-AI's evolution requires education systems to abandon binary "ban or embrace", **focusing instead on integration**. For AI development, growing weight of *AI Tool Proficiency* highlights the need to prioritize user-centric practicality over pure technical advancement. For higher education, STEM programs should deepen AI as a tool, trade schools should augment hands-on skills with AI utilities, and arts institutions should safeguard human originality.

Overall, our study demonstrates that adapting to Gen-AI requires aligning educational strategies with career-specific impacts, supported by dynamic, data-driven frameworks.

Reference & Resources

- [1] K. Pathak, G. Prakash, A. Samadhiya, A. Kumar, S. Luthra, Impact of Gen-AI chatbots on consumer services experiences and behaviors: Focusing on the sensation of awe and usage intentions through a cybernetic lens, *J. Retail. Consum. Serv.* 82 (2025) 104120. <https://doi.org/10.1016/j.jretconser.2024.104120>.
- [2] W. Choi, The role of changes in pronunciation ability and anxiety through Gen-AI on EFL learners' self-directed speaking motivation and social interaction confidence - A CHAT perspective, *System* 133 (2025) 103788. <https://doi.org/10.1016/j.system.2025.103788>.
- [3] L. Fontoura, D. Luiz de Mattos Nascimento, J.V. Neto, R.G. Gusmão Caiado, Energy Gen-AI technology framework: A perspective of energy efficiency and business ethics in operation management, *Technol. Soc.* 81 (2025) 102847. <https://doi.org/10.1016/j.techsoc.2025.102847>.
- [4] D. Teng, C. Ye, V. Martinez, Gen-AI's effects on new value propositions in business model innovation: Evidence from information technology industry, *Technovation* 143 (2025) 103191. <https://doi.org/10.1016/j.technovation.2025.103191>.
- [5] T. Yordudom, C. Nui-Suk, N. Imjai, S. Sumiati, S. Aujiirapongpan, Developing open innovation competency in Gen Z entrepreneurship students: Roles of absorptive capacity, AI literacy and algorithmic thinking skills, *Int. J. Manag. Educ.* 24 (2026) 101326. <https://doi.org/10.1016/j.ijme.2025.101326>.
- [6] S. Swami, J. Banhazi, H. Irfan, T. Srivastava, MSR153 Frequently Asked Questions (FAQs): What Are the Risks and Mitigation Strategies in Using Gen-AI in HEOR?, *Value Health* 28 (2025) S304. <https://doi.org/10.1016/j.jval.2025.04.1304>.
- [7] A.A. Funa, R.A.E. Gabay, Exploring perspectives toward generative artificial intelligence integration in science education: A cross-generational study, *Soc. Sci. Humanit. Open* 13 (2026) 102358. <https://doi.org/10.1016/j.ssaho.2025.102358>.
- [8] X. Zhou, Z. Zhang, X. Lu, J. Chen, System dynamics-based dynamic evaluation of value synergy and effectiveness verification of governance strategies for urban water conservancy projects, *Expert Syst. Appl.* 309 (2026) 131208. <https://doi.org/10.1016/j.eswa.2026.131208>.
- [9] Yu Jianxin; Cheng Minqian; Xu Tao, An Enterprise Data Asset Quality Evaluation Model Based on Cloud Model, *J. Tongji Univ. Nat. Sci. Ed.* YR 2026 (n.d.) 314–322.
- [10] Liu Xinqi; Qi Wei, Risk Assessment of Multi-modal Transport in Mountain Scenic Areas Based on Combined Weighting - Cloud Model, *Logist. Technol.* YR 2026 (n.d.) 103–110.
- [11] <https://www.weforum.org/> World Economic Forum, WEF.
- [12] <https://www.ieee.org/> IEEE Data of engineering career (STEM).
- [13] <https://www.dol.gov/agencies/eta/research/publications/voluntary-skill-standards-and-certification> Skills Resources.
- [14] <https://www.insea.org/> International Society for Education Through Art (InSEA).

Report on Use of AI Tool

Tool Used: Gemini Pro

Purpose: Structuring Methodology, Helping Defining Variables, and Part of Coding.

Conversation 1: Methodology Framework

Query : I am developing a framework that combines Entropy Weight-Cloud TOPSIS and System Dynamics for the 2026 MCM. Can you help me find a professional name for this hybrid model and describe its core logic in one concise paragraph?

Output : You could call it the Hybrid Evaluation-Evolution Framework (HEEF). This model functions by coupling a static multi-criteria evaluation (Entropy Weight-Cloud TOPSIS) with a long-term temporal simulation (System Dynamics). The evaluation results serve as dynamic inputs to modify the evolutionary parameters, capturing the non-linear adaptation of the system.

Accept : We established a hybrid Evaluation-Evolution Framework coupling Entropy Weight-Cloud TOPSIS with System Dynamics (SD). ... Crucially, a closed-loop feedback mechanism is introduced where evaluation results dynamically modify evolutionary parameters, ensuring that the model captures the non-linear adaptation of educational systems.

Conversation 2: Enhancing the Summary Professionalism

Query : Can you provide a powerful opening quote for a summary sheet about Generative AI, education, and career development? It should emphasize adaptation and change.

Output : A suitable quote would be: 'When the winds of change blow, some build walls while others build windmills.' This highlights the proactive stance your paper takes on Gen-AI integration.

Accept : "When the winds of change blow, some build walls while others build windmills." As Generative AI (Gen-AI) shifts from a novelty to a fundamental technical force, aligning higher education with evolving career demands is significant.

Conversation 3: Describing Simulation Results for Specific Careers

Query : Based on my simulation results, the Data Scientist (STEM) has a rising relative closeness from 0.7683 to 0.7764. How can I describe this trend in a professional academic way?

Output : The simulation reveals that the STEM trajectory, specifically for the Data Scientist role, demonstrates superior resilience. The relative closeness value increases from 0.7683 in 2026 to 0.7764 in 2040, a trend primarily driven by a high degree of AI synergy within the field.

Accept: The simulation reveals distinct adaptive trajectories. STEM (Data Scientist) demonstrates superior resilience with relative closeness rising from 0.7683 (2026) to 0.7764 (2040), driven by high AI synergy.

Conversation 4: Picture Drawing

Query : Could you please help me create a career chart based on the three professions of STEM, Trade and Arts, as well as their future plans?

Output : According to your request, we will create a scientific cartoon-style illustration.

Accept: Fig. 1 illustration.